



12.企业SQL面试题

题目1：腾X面试题（留存、topN）

用HiveSQL实现

(1) 全量用户登录日志表 `t_login_all`，字段信息 `ftime`（登录日期）、`openid`（登录帐号）

新增用户登录日志表 `t_login_new`，字段信息 `ftime`（登录日期）、`openid`（登录帐号）

求每天新增用户次日、7天、30天留存率。

(说明：7天留存是指当天有登录且第7天还登录的用户)

分析：留存问题

```
select
  t3.new_ftime
, count(case when t3.diff=1 then t3.openid end)/count(t3.openid) as 1day_user_cnt
, count(case when t3.diff=6 then t3.openid end)/count(t3.openid) as 7day_user_cnt
, count(case when t3.diff=29 then t3.openid end)/count(t3.openid) as 30day_user_cnt
from
(
  select t1.openid
    , t1.ftime new_ftime
    , t2.ftime all_ftime
    , datediff(t1.ftime, t2.ftime) as diff
  from (select distinct ftime, openid from ds_hive.ch13_t_login_new) t1
 left join (select distinct ftime, openid from ds_hive.ch13_t_login_all) t2
    on t1.openid=t2.openid
 where datediff(t1.ftime, t2.ftime) <= 30
) t3
group by t3.new_ftime
;
```

运行结果：

第一小题：

	t3.new_ftime	1day_user_cnt	7day_user_cnt	30day_user_cnt
1	2025-06-01	0	0	0
2	2025-06-02	0.3333333333333333	0	0
3	2025-06-07	0	0.3	0
4	2025-06-30	0	0	0.25

(2) 消息流水表 t_chat_all ,

字段信息: ftime (日期)、 send_user_id (发消息用户id)、 receive_user_id (接收消息用户id)、 chat_id (消息id)、 send_time (发消息时间)

用户登录流水日志表 t_user_login_all ,

字段信息: ftime (日期)、 user_id (用户id)、 login_id (登录id)、 login_loc (登录区服)、 login_time (登录时间)

求: 有收发消息且当天登录的用户中, 最近登录时间、登录区服, 输出 ftime , user_id , login_loc , login_time

第二小题:

查询历史记录 保存的查询 查询生成器 结果 (3)

	t4.ftime	t4.user_id	t4.login_loc	t4.login_time
1	2025-06-01	1001	01	2025-06-01 12:34:45
2	2025-06-01	1002	01	2025-06-01 09:34:45
3	2025-06-01	1003	01	2025-06-01 11:34:45

分析: topN问题, 窗口函数

```

-- timu1_user_login有收发消息且当天登录的所有用户记录
with timu1_user_login as(
select
    t2.user_id
    ,t2.ftime
    ,t3.login_loc
    ,t3.login_time
from
    (
        select  distinct
            user_id
            ,ftime
        from
            (
                select send_user_id as user_id,ftime  from ds_hive.ch13_t_chat_all
                union all
                select receive_user_id as user_id,ftime  from ds_hive.ch13_t_chat_all
            ) t1
    ) t2 --当天所有活跃（有收发消息）的用户列表
join
    (
        select distinct ftime,user_id,login_loc,login_time from ds_hive.ch13_t_user_login_all
    ) t3
on t2.user_id=t3.user_id and t2.ftime=t3.ftime
)

select  t4.ftime
    ,t4.user_id
    ,t4.login_loc
    ,t4.login_time
from
    (
        select
            ftime
            ,user_id
            ,login_loc
            ,login_time
            ,row_number() over(partition by user_id order by ftime desc ) as num
        from timu1_user_login

```

```
) t4
where t4.num=1
;
```

运行结果：

查询历史记录

保存的查询

查询生成器

结果 (1)

uid_cnt

1	2
---	---

题目2：快X面试题(json解析、谓词下推)

表： user:uid,age,date

order:order_id,order_money,location,date , 其中 location:{"city","xx"}

双十一场景，找出北京市双十一（11.1~11.12），年龄范围在20-25岁的人中的人数，人均销售额大于1000的人数

分析：json解析、谓词下推

```

select count(t3.uid)    as uid_cnt
from
(select t1.uid
      ,get_json_object(t1.location,'$.city') as city
      ,t2.age
      ,sum(t1.order_money)                  as sum_order_money
  from ds_hive.ch13_timu2_order t1
 left join ds_hive.ch13_timu2_user t2
    on t1.uid=t2.uid and t2.age>=20 and t2.age<=25
 where t1.order_date>='2025-11-01' and t1.order_date<='2025-11-12'
 group by t1.uid
        ,get_json_object(t1.location,'$.city')
        ,t2.age
 having sum_order_money>1000
 ) t3
;

```

运行结果：

查询历史记录

保存的查询

查询生成器

结果 (1)

uid_cnt

1 2

题目3：字X面试题（连续问题）

有一张用户签到表【ds_hive.ch13_user_attendence】，标记每天用户是否签到

（说明：该表包含所有用户所有工作日的出勤记录），包含三个字段：

日期【fdate】

用户id【fuser_id】

用户当天是否签到【fis_sign_in：0否1是】

问题1：请计算截至当前每个用户已经连续签到的天数（输出表仅包含当天签到的所有用户，计算其连续签到天数）

问题2：请计算每个用户历史以来最大的连续签到天数（输出表为用户签到表中所有出现过的用户，计算其历史最大连续签到天数）

分析：连续问题，只有工作日需要记录

```
-- 问题1
select
    t1.fuser_id
    ,(case when datediff t2.fdate is null and t3.fdate is not null then '满勤'
        when datediff t2.fdate is null then '1'
        else datediff(t1.fdate,t2.fdate) end) as days
from
(
    select fuser_id
        ,fdate
        from ds_hive.ch13_user_attendance
        where fdate=current_date()
        and fis_sign_in='1'
) t1
left join
(
    select fuser_id
        ,max(fdate) as fdate
        from ds_hive.ch13_user_attendance
        where fdate<=current_date()
        and fis_sign_in='0'
        group by fuser_id
)
    t2
    on t1.fuser_id=t2.fuser_id
left join
(
    select fuser_id
        ,min(fdate) as fdate
        from ds_hive.ch13_user_attendance
        where fdate<=current_date()
        and fis_sign_in='1'
        group by fuser_id
) t3
    on t1.fuser_id=t3.fuser_id
;
```

-- 问题1 方法二:

```
with timu_user_attendance_tmp3 as (  
select t1.fuser_id  
      ,t1.rank_dt  
      ,count(*) as cnt  
from  
  (  
    SELECT  
      fuser_id,  
      fdate,  
      fis_sign_in,  
      date_add(fdate,ROW_NUMBER() OVER (PARTITION BY fuser_id ORDER BY fdate desc)) AS rank_dt  
    FROM  
      ds_hive.ch13_user_attendance  
    WHERE  
      fis_sign_in = 1  
  ) t1  
group by t1.fuser_id,t1.rank_dt  
)  
  
select fuser_id  
      ,cnt  
from timu_user_attendance_tmp3  
where rank_dt=date_add(current_date(),1)  
;
```

```
-- 问题2
with timu_user_attendance_tmp4 as (
select t1.fuser_id
      ,t1.rank_dt
      ,count(*) as cnt
from
  (
    SELECT
      fuser_id,
      fdate,
      fis_sign_in,
      date_add(fdate,ROW_NUMBER() OVER (PARTITION BY fuser_id ORDER BY fdate desc)) AS rank_dt
    FROM
      ds_hive.ch13_user_attendance
    WHERE
      fis_sign_in = 1
  ) t1
group by t1.fuser_id,t1.rank_dt
)

select  fuser_id
      ,max(cnt) as max_cnt
from timu_user_attendance_tmp4
group by fuser_id;
```

运行结果

第一小题:

查询历史记录 		保存的查询	查询生成器	结果 (2)
t1.fuser_id		days		
1	002	满勤		
2	003	1		

第二小题:

	fuser_id	cnt
1	002	3
2	003	1
3	004	1

题目4：华X面试题（列转行）

有一个录取学生人数表，记录的是每年录取学生人数和入学学生的学制以下是表结构：

```
CREATE TABLE admit(
  id int(11) ,
  year int(255)          COMENT '入学年度',
  num int(255)           COMENT '录取学生人数',
  stu_len varchar(255)  CONENT '学生学制'
);
```

`year` 表示学生入学年度，`num` 表示对应年度录取人数，`stu_len` 表示录取学生的学制
说明: 例如录取年度2018学制3，表示该批学生在校年份为2018 ~ 2019、2019 ~ 2020,2020 ~ 2021

id	year	num	stu_len
1	2018	2000	3
2	2019	2000	3
3	2020	1000	4
4	2021	2000	3

根据以上示例计算出每年在校人数,写出SQL语句:

分析：列转行、爆炸函数

```

with timu4_stu_tmp1 as(
select
id
,year
,num
,stu_len
,space(cast(stu_len-1 as int))
,split(space(cast(stu_len-1 as int)),'') as dt_year
from ds_hive.ch13_t_stu_school
),
timu4_stu_tmp2 as(
select  id
        ,year
        ,num
        ,stu_len
        ,`空字符串`
        ,row_number() over(partition by id order by year) as rank
    from timu4_stu_tmp1
    lateral view explode(dt_year) ex as `空字符串`
)
select year+rank-1    as tear
        ,sum(num) as stu_cnt
    from timu4_stu_tmp2
group by year+rank-1;

```

LATERAL VIEW explode(数组字段) 虚拟表别名 AS 列别名

运行结果：

timu4_stu_tmp1:

	id	year	num	stu_len	_c4	dt_year
1	1	2018	2000	3		["", "", ""]
2	2	2019	2000	3		["", "", ""]
3	3	2020	1000	4		["", "", "", ""]
4	4	2021	2000	3		["", "", ""]

	id	year	num	stu_len	空字符串列	rank
1	1	2018	2000	3		1
2	1	2018	2000	3		2
3	1	2018	2000	3		3
4	2	2019	2000	3		1
5	2	2019	2000	3		2
6	2	2019	2000	3		3
7	3	2020	1000	4		1
8	3	2020	1000	4		2
9	3	2020	1000	4		3
10	3	2020	1000	4		4

tear	stu_cnt
2018	2000
2019	4000
2020	5000
2021	5000
2022	3000
2023	3000

题目5：字X面试题（性能优化）

有一张员工入职时间表 `ds_hive.ch13_t_new_emp`，如下：

	ch13_t_new_emp.employee_id	ch13_t_new_emp.department_id	ch13_t_new_emp.hire_date
1	1001	a	2024-11-16
2	1002	a	2024-11-25
3	1003	a	2024-11-30
4	1004	a	2024-12-01
5	1005	a	2024-12-02
6	1006	a	2024-10-01
7	1007	a	2024-10-02

要求：按照部门已入职员工的入职时间排序，给每位员工编号。

```
select
    employee_id
    ,department_id
    ,hire_date
    ,row_number() over(partition by department_id order by hire_date ) as rn
from ds_hive.ch13_t_new_emp
```

	employee_id	department_id	hire_date	rn
1	1006	a	2024-10-01	1
2	1007	a	2024-10-02	2
3	1001	a	2024-11-16	3
4	1002	a	2024-11-25	4
5	1003	a	2024-11-30	5
6	1004	a	2024-12-01	6
7	1005	a	2024-12-02	7

-----优化后的sql

```

with ch13_t_new_emp_tmp1 as (
    select
employee_id
,department_id
,hire_date
,substr(hire_date,1,7) as months
,row_number() over(partition by department_id,substr(hire_date,1,7) order by hire_date) as num
from ds_hive.ch13_t_new_emp
),
ch13_t_new_emp_tmp2 as(
    select
        t2.department_id
    ,t2.months
    ,lag(t2.cnt,1,0) over(partition by t2.department_id order by t2.months) lag_cnt
from
(
    select
        t1.department_id
    ,t1.months
    ,sum(t1.cnt) over(partition by t1.department_id order by t1.months rows between unbounded
from
(
    select
        department_id
        ,substr(hire_date,1,7) as months
        ,count(employee_id) as cnt
    from ds_hive.ch13_t_new_emp
    group by department_id
        ,substr(hire_date,1,7)
    ) t1
    )t2
)
select t1.employee_id
        ,t2.department_id
        ,t1.hire_date
        ,num+lag_cnt as num
from ch13_t_new_emp_tmp1 t1
left join ch13_t_new_emp_tmp2 t2
on t1.department_id=t2.department_id

```

```
and t1.months=t2.months
```

```
;
```